

# Rahul Rawat

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## Freudenberg Technology Innovation

Hiring Team

### Application: Master's Thesis in Data Science and Machine Learning in Injection Molding in Weinheim

Dear Hiring Team,

I am applying for the Master's Thesis in the Field of Data Science and Machine Learning in Injection Molding at Freudenberg Technology Innovation in Weinheim. The brief on analyzing and preparing production and process data from injection molding systems, identifying influencing factors on part quality, cycle time, and scrap rates, developing regression, classification, and anomaly detection models, and translating results into actionable process optimization recommendations for technical stakeholders, maps directly to the work I have been shipping over the past year.

At eRay GmbH I built an end to end recursive time series pipeline forecasting four water quality indicators, chlorophyll a, turbidity, pH, and dissolved oxygen, and benchmarked six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, using CatBoost multi quantile regression to produce asymmetric 80 percent prediction intervals for decision support under uncertainty. I enforced strict anti leakage rules across the pipeline and surfaced the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors. That leakage aware and honest evaluation mindset is exactly what a Master's Thesis on injection molding process data needs, where cycle time, part quality, and scrap rates all depend on getting the evaluation right rather than reading a plausible looking accuracy number.

In my Movie Analytics and ML Pipeline on GCP I trained a leakage free BigQuery ML classifier that predicts a target before release, deliberately splitting features into two tables so only pre release signals feed the model, and delivered a five page Looker Studio dashboard for concrete business questions. In my Real Time Flight Tracking Pipeline I processed over 128 thousand records with PySpark on Google Cloud, collecting live positions from an API every 30 seconds and joining them against airport, aircraft, and weather data across four sources with dbt, orchestrated by Apache Airflow so batch and real time layers refresh every 15 minutes. Both projects show I can build the ingestion, preparation, modelling, and visualization layers a manufacturing analytics thesis needs, end to end.

I am proficient in Python and comfortable with pandas, scikit learn, and PyTorch, and I hold the AWS Academy Cloud Foundations, SAS Certified Specialist Visual Business Analytics Using SAS Viya, and Google Data Analytics Foundations certificates. My English is fluent working proficiency and my current German level is B1 in progress toward B2, and I meet the very good German or English requirement in English. I am interested in continuing into an industrial PhD after the thesis and would be glad to align the exact scope of the injection molding data set and evaluation criteria with the Weinheim team.

Kind regards,

**Rahul Rawat**